

PMI Model Specification

Akaike Info Criterion (AIC) - HAC (Newey-West)

lnpmi ln_cif ln_customs_duties_taxes ln_opex ir ln_fx

m2 m3 m4 m5 m6 m7 m8 m9 m10 m11 m12 p2020m03 p2020m04 p2020m05
p2020m06 p2020m07 p2020m08 p2022m03

CASE 3

Bounds Tests

Null hypothesis: No levels relationship
Number of dynamic cointegrating variables: 5
Deterministics: Unrest. constant (Case 3)
Sample size: 176

Test Statistic	Value
F-statistic	7.198931
t-statistic	-5.056502

Bounds Critical Values

Sample Size		10%	5%	1%
F-Statistic				
I(0)	Asymptotic	2.260	2.620	3.410
I(1)	Asymptotic	3.350	3.790	4.680
t-Statistic				
I(0)	Asymptotic	-2.570	-2.860	-3.430
I(1)	Asymptotic	-3.860	-4.190	-4.790

* I(0) and I(1) are respectively the stationary and non-stationary bounds.

Conditional Error Correction

Dependent Variable: LNPMI

Method: ARDL

Date: 09/09/26 Time: 09:52

Sample: 2011M05 2025M12

Included observations: 176

Lag selection: Automatic (deplags=4, reglags=4)

Selected model: ARDL(4,0,0,4,0,2) using AIC (12500 models evaluated)

HAC standard errors & covariance (Bartlett kernel, Newey-West fixed bandwidth = 5.0000)

Variable	Coefficient	Std. Error	t-Statistic	Prob.*
Long-run Regressors				
Linear: Dependent				
LNPMI(-1)	-0.381973	0.075541	-5.056502	0.0000
Linear: Independent				
LN_CIF	-0.019459	0.008210	-2.370270	0.0191
LN_CUSTOMS_DUTIES_TAX...	-0.035193	0.022448	-1.567746	0.1192
LN_OPEX(-1)	0.013857	0.011420	1.213384	0.2270
IR	-0.005215	0.003849	-1.354988	0.1776
LN_FX(-1)	0.047230	0.080156	0.589225	0.5566
Short-run Regressors				

Linear: Dependent				
D(LNPMI(-1))	-0.262211	0.084414	-3.106248	0.0023
D(LNPMI(-2))	-0.165569	0.056320	-2.939809	0.0038
D(LNPMI(-3))	-0.116704	0.070048	-1.666062	0.0979
Linear: Independent				
D(LN_OPEX)	-0.002608	0.014069	-0.185371	0.8532
D(LN_OPEX(-1))	0.012423	0.015453	0.803935	0.4228
D(LN_OPEX(-2))	0.020448	0.015126	1.351832	0.1786
D(LN_OPEX(-3))	0.033473	0.011087	3.019066	0.0030
D(LN_FX)	0.048134	0.120355	0.399933	0.6898
D(LN_FX(-1))	-0.354642	0.161560	-2.195105	0.0298
Deterministic				
C	2.069146	0.416950	4.962581	0.0000
Fixed				
M2	0.002025	0.016143	0.125446	0.9003
M3	-0.000102	0.014470	-0.007042	0.9944
M4	0.017917	0.013977	1.281942	0.2020
M5	0.006440	0.015609	0.412555	0.6806
M6	0.005392	0.012896	0.418129	0.6765
M7	-0.005756	0.014986	-0.384071	0.7015
M8	0.003585	0.014621	0.245229	0.8066
M9	0.005551	0.012966	0.428141	0.6692
M10	0.020023	0.012576	1.592110	0.1136
M11	0.008266	0.013310	0.621038	0.5356
M12	0.025540	0.017569	1.453680	0.1482
P2020M03	-0.667098	0.010392	-64.19567	0.0000
P2020M04	-0.450467	0.055769	-8.077364	0.0000
P2020M05	-0.101714	0.055760	-1.824132	0.0702

P2020M06	0.069585	0.060528	1.149633	0.2522
P2020M07	0.109993	0.034303	3.206493	0.0017
P2020M08	-0.029678	0.029370	-1.010463	0.3140
P2022M03	0.173109	0.014848	11.65879	0.0000
R-squared	0.851209	Mean dependent var	-0.001054	
Adjusted R-squared	0.816631	S.D. dependent var	0.068688	
S.E. of regression	0.029413	Akaike info criterion	-4.043025	
Sum squared resid	0.122851	Schwarz criterion	-3.430545	
Log likelihood	389.7862	Hannan-Quinn criter.	-3.794606	
F-statistic	24.61697	Durbin-Watson stat	2.081221	
Prob(F-statistic)	0.000000			

☐ Error Correction

Dependent Variable: LNPMI

Method: ARDL

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Lag selection: Automatic (deplags=4, relags=4)

Selected model: ARDL(4,0,0,4,0,2) using AIC (12500 models evaluated)

HAC standard errors & covariance (Bartlett kernel, Newey-West fixed bandwidth = 5.0000)

Variable	Coefficient	Std. Error	t-Statistic	Prob.*
Cointegrating Equation				
COINTEQ	-0.381973	0.059823	-6.384998	0.0000
Short-run Regressors				
<u>Linear: Dependent</u>				
D(LNPMI(-1))	-0.262211	0.071819	-3.650972	0.0004
D(LNPMI(-2))	-0.165569	0.052270	-3.167565	0.0019
D(LNPMI(-3))	-0.116704	0.068673	-1.699405	0.0914
<u>Linear: Independent</u>				
D(LN_OPEX)	-0.002608	0.013820	-0.188712	0.8506
D(LN_OPEX(-1))	0.012423	0.014644	0.848340	0.3976
D(LN_OPEX(-2))	0.020448	0.015256	1.340371	0.1822
D(LN_OPEX(-3))	0.033473	0.011186	2.992524	0.0032
D(LN_FX)	0.048134	0.118683	0.405566	0.6857

Fixed				
M2	0.002025	0.015105	0.134061	0.8935
M3	-0.000102	0.014401	-0.007076	0.9944
M4	0.017917	0.013481	1.329055	0.1859
M5	0.006440	0.015150	0.425067	0.6714
M6	0.005392	0.012490	0.431718	0.6666
M7	-0.005756	0.014603	-0.394146	0.6940
M8	0.003585	0.014246	0.251689	0.8016
M9	0.005551	0.012516	0.443519	0.6580
M10	0.020023	0.012304	1.627298	0.1058
M11	0.008266	0.012998	0.635930	0.5258
M12	0.025540	0.016929	1.508711	0.1335
P2020M03	-0.667098	0.010247	-65.10362	0.0000
P2020M04	-0.450467	0.053283	-8.454305	0.0000
P2020M05	-0.101714	0.053165	-1.913156	0.0577
P2020M06	0.069585	0.059303	1.173389	0.2425
P2020M07	0.109993	0.031238	3.521175	0.0006
P2020M08	-0.029678	0.029151	-1.018076	0.3103
P2022M03	0.173109	0.012327	14.04255	0.0000
R-squared	0.851209	Mean dependent var	-0.001054	
Adjusted R-squared	0.822868	S.D. dependent var	0.068688	
S.E. of regression	0.028909	Akaike info criterion	-4.099843	
Sum squared resid	0.122851	Schwarz criterion	-3.577433	
Log likelihood	389.7862	Hannan-Quinn criter.	-3.887956	
F-statistic	30.03443	Durbin-Watson stat	2.081221	
Prob(F-statistic)	0.000000			

☐ Cointegrating Specification

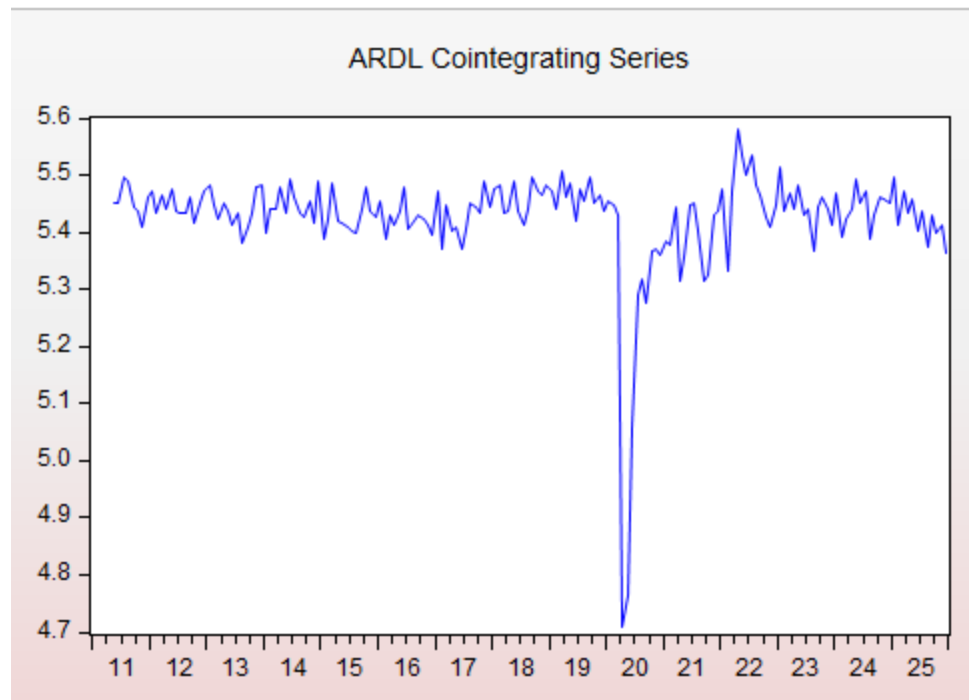
Deterministics: Unrest. constant (Case 3)

$$CE = LNPMI(-1) - (-0.050944 * LN_CIF - 0.092135 * LN_CUSTOMS_DUTIES_TAXES + 0.036276 * LN_OPEX(-1) - 0.013653 * IR + 0.123648 * LN_FX(-1))$$

☐ Cointegrating Coefficients

Deterministics: Unrest. constant (Case 3)

Variable	Coefficient	Std. Error	t-Statistic	Prob.
LN_CIF	-0.050944	0.018035	-2.824774	0.0053
LN_CUSTOMS_DUTIES_TAX...	-0.092135	0.057430	-1.604287	0.1105
LN_OPEX(-1)	0.036276	0.025549	1.419893	0.1575
IR	-0.013653	0.010201	-1.338380	0.1825
LN_FX(-1)	0.123648	0.209705	0.589627	0.5562



Ramsey RESET Test

Equation: UNTITLED

Omitted Variables: Squares of fitted values

Specification: LNPMI LNPMI(-1) LNPMI(-2) LNPMI(-3) LNPMI(-4) LN_CIF
 LN_CUSTOMS_DUTIES_TAXES LN_OPEX LN_OPEX(-1)
 LN_OPEX(-2) LN_OPEX(-3) LN_OPEX(-4) IR LN_FX LN_FX(-1)
 LN_FX(-2) C M2 M3 M4 M5 M6 M7 M8 M9 M10 M11 M12
 P2020M03 P2020M04 P2020M05 P2020M06 P2020M07
 P2020M08 P2022M03

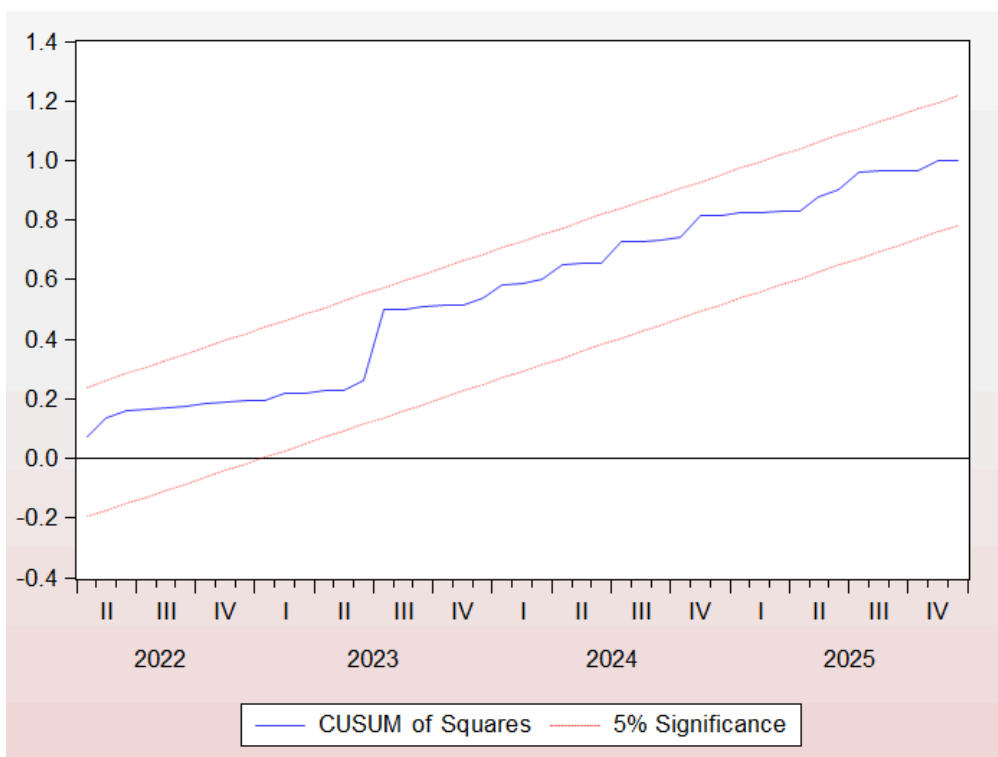
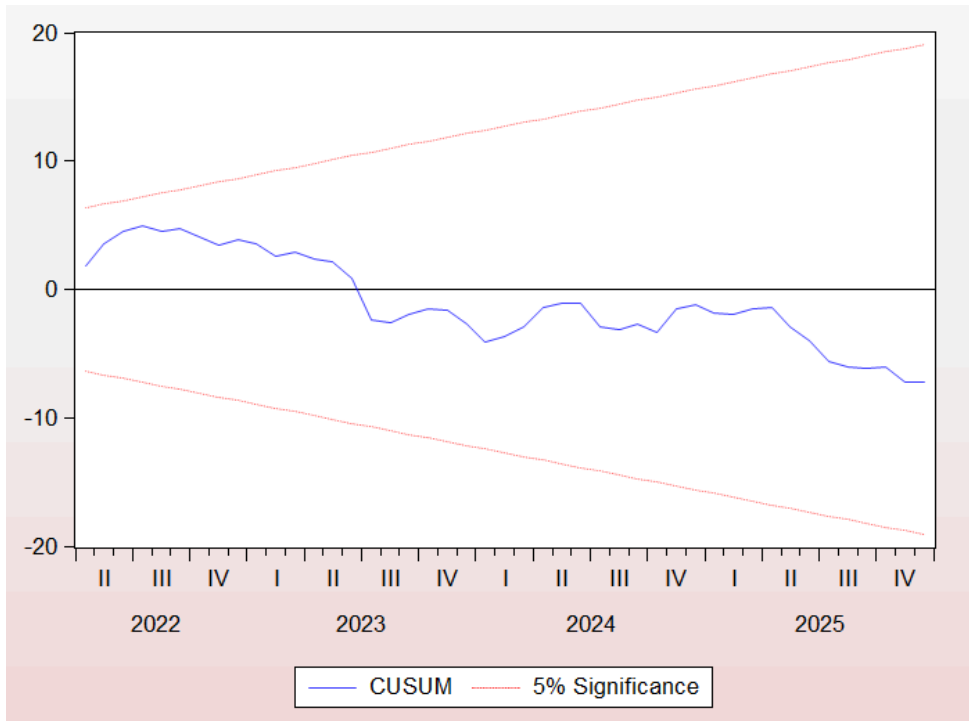
	Value	df	Probability
t-statistic	1.399411	141	0.1639
F-statistic	1.958350	(1, 141)	0.1639
Likelihood ratio	2.427646	1	0.1192

F-test summary:

	Sum of Sq.	df	Mean Squares
Test SSR	0.001683	1	0.001683
Restricted SSR	0.122851	142	0.000865
Unrestricted SSR	0.121168	141	0.000859

LR test summary:

	Value
Restricted LogL	389.7862
Unrestricted LogL	391.0000



Heteroskedasticity Test: Breusch-Pagan-Godfrey

Null hypothesis: Homoskedasticity

F-statistic	1.043622	Prob. F(33,142)	0.4156
Obs*R-squared	34.35373	Prob. Chi-Square(33)	0.4027
Scaled explained SS	33.64491	Prob. Chi-Square(33)	0.4361

Breusch-Godfrey Serial Correlation LM Test:

Null hypothesis: No serial correlation at up to 2 lags

F-statistic	1.870134	Prob. F(2,140)	0.1579
Obs*R-squared	4.579698	Prob. Chi-Square(2)	0.1013

